

NE 470/570 Monte Carlo Methods for Radiation Transport

Topic 3: Fundamentals of Probability and Statistics

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Outline

1. Introduction
2. Expectation
3. Sample Statistics
4. Common Density Functions
5. Limit Theorems
6. Relative Uncertainty
7. Confidence Levels
8. Normality Tests

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■ **Statistics vs. Probability**

- 1 “The problems considered by probability and statistics are **inverse** to each other. In **probability** theory we consider some underlying process which has some randomness or uncertainty modeled by random variables, and we figure out what happens. In **statistics** we observe something that has happened, and try to figure out what underlying process would explain those observations.” - Persi Diaconis
- 2 “**Probability** deals with predicting the likelihood of **future** events, while **statistics** involves the analysis of the frequency of **past** events. Probability is primarily a **theoretical** branch of mathematics, which studies the consequences of mathematical definitions. Statistics is primarily an **applied** branch of mathematics, which tries to make sense of observations in the real world.” - Steven Skiena

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- **Statistics vs. Probability**

- 1 *"The problems considered by probability and statistics are **inverse** to each other. In **probability** theory we consider some underlying process which has some randomness or uncertainty modeled by random variables, and we figure out what happens. In **statistics** we observe something that has happened, and try to figure out what underlying process would explain those observations."* - Persi Diaconis
 - 2 *"**Probability** deals with predicting the likelihood of **future** events, while **statistics** involves the analysis of the frequency of **past** events. Probability is primarily a **theoretical** branch of mathematics, which studies the consequences of mathematical definitions. Statistics is primarily an **applied** branch of mathematics, which tries to make sense of observations in the real world."* - Steven Skiena
- The **MC approach is a statistical method** that utilizes RNs to sample an **unknown population**, e.g., particle history, and consequently, evaluates the expected outcome of a physical process.
 - To present the results of a statistical analysis, we will estimate the following quantities:
 - 1 The **sample mean/average**
 - 2 The **sample variance/standard deviation** and **relative uncertainty**
 - 3 The **confidence level**

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2. Expectation

2.1. Single Random Variable

2.2. Multiple Random Variables

- For a **continuous** RV X , whose PDF is $p(x)$ in a range of $[a, b]$. The **expectation value** of a function $g(X)$ is:

$$\mathbb{E}[g(X)] = \int_a^b g(x)p(x)dx \quad (1)$$

- For a **discrete** RV X that has N outcomes $\{x_i\}_{i=1}^N$, the expectation value of $g(X)$ is:

$$\mathbb{E}[g(X)] = \sum_{i=1}^N g(x_i)p(x_i) \quad (2)$$

- If $g(x) = x$, $\mathbb{E}[g(X)]$ is simply called the **expectation** or the **true mean**.
- For a **continuous** RV X :

$$m_x = \mathbb{E}[X] = \int_a^b xp(x)dx \quad (3)$$

- For a **discrete** RV X :

$$m_x = \mathbb{E}[X] = \sum_{i=1}^N x_i p(x_i) \quad (4)$$

- The k^{th} moment for a **continuous** RV X is defined as:

$$m_x^k = \mathbb{E} [X^k] = \int_a^b x^k p(x) dx \quad (5)$$

- The k^{th} moment for a **discrete** RV X is defined as:

$$m_x^k = \mathbb{E} [X^k] = \sum_{i=1}^N x_i^k p(x_i) \quad (6)$$

- The k^{th} **moment** for a **continuous** RV X is defined as:

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- The k^{th} **moment** for a **discrete** RV X is defined as:

$$m_x^k = \mathbb{E} [X^k] = \sum_{i=1}^N x_i^k p(x_i) \quad (6)$$

- The k^{th} **central moment** for a **continuous** RV X is defined as:

$$\mathbb{E} [(X - m_x)^k] = \int_a^b (x - m_x)^k p(x) dx \quad (7)$$

- The k^{th} **central moment** for a **discrete** RV X is defined as:

$$\mathbb{E} [(X - m_x)^k] = \sum_{i=1}^N (x_i - m_x)^k p(x_i) \quad (8)$$

- The central moment of order 2 ($k = 2$) is referred to as the **true variance** of X .

- For a **continuous** RV:

$$\begin{aligned}\text{Var}[X] &= \sigma_x^2 = \mathbb{E}[(X - m_x)^2] \\ &= \int_a^b (x - m_x)^2 p(x) dx\end{aligned}\tag{9}$$

- For a **discrete** RV:

$$\begin{aligned}\text{Var}[X] &= \sigma_x^2 = \mathbb{E}[(X - m_x)^2] \\ &= \sum_{i=1}^N (x_i - m_x)^2 p(x_i)\end{aligned}\tag{10}$$

- An important relation:

$$\begin{aligned}\sigma_x^2 &= \int_a^b (x^2 + m_x^2 - 2xm_x)p(x)dx \\ &= \mathbb{E}[X^2] + m_x^2 - 2m_x^2 \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2\end{aligned}\tag{11}$$

- The square root of the variance is referred to as the **true standard deviation** $\sigma_x = \sqrt{\sigma_x^2}$.
- The standard deviation is an indication of the **dispersion** of RV X relative to its mean (m_x).
- The **true mean** and **true variance** are also referred to as the **population parameters**, because they are obtained based on a **known PDF**, i.e., **population**.

Useful formulation for the expectation operator

- Expectation of a RV or a function of a RV multiplied by a constant.

$$\mathbb{E}[ag(X) + b] = a\mathbb{E}[g(X)] + b \quad (12)$$

- Expectation of a linear combination of a few RVs or functions of RVs.

$$\mathbb{E}\left[\sum_{k=1}^K a_k g_k(X_k)\right] = \sum_{k=1}^K a_k \mathbb{E}[g_k(X_k)] \quad (13)$$

- Variance of a RV or a function of a RV multiplied by a constant.

$$\text{Var}[ag(X) + b] = a^2 \text{Var}[g(X)] \quad (14)$$

- Variance of a linear combination of a few RVs or functions of RVs.

$$\text{Var}\left[\sum_{k=1}^K a_k g_k(X_k)\right] = \sum_{k=1}^K a_k^2 \text{Var}[g_k(X_k)] + \sum_{1 \leq i < j \leq K} 2a_i a_j \text{Cov}(g_i(X_i), g_j(X_j)) \quad (15)$$

where $\text{Cov}(X, Y)$ is the **covariance** defined as

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \quad (16)$$

Outline

2. Expectation

2.1. Single Random Variable

2.2. Multiple Random Variables

- Suppose a RV is made of a combination of other RVs, such as:

$$X_3 = c_1 X_1 + c_2 X_2 \quad (17)$$

- The variance of X_3 can be derived as below:

$$\begin{aligned} \text{Var}[X_3] &= \text{Var}[c_1 X_1 + c_2 X_2] \\ &= \mathbb{E} \left[(c_1 X_1 + c_2 X_2 - c_1 m_{x_1} - c_2 m_{x_2})^2 \right] \\ &= c_1^2 \sigma_{x_1}^2 + c_2^2 \sigma_{x_2}^2 + 2c_1 c_2 \text{Cov}(X_1, X_2) \end{aligned} \quad (18)$$

where $\text{Cov}(X_1, X_2) = \mathbb{E}[(X_1 - m_{x_1})(X_2 - m_{x_2})]$.

- The **correlation coefficient** between two RVs:

$$\rho_{X_1, X_2} = \frac{\text{Cov}(X_1, X_2)}{\sigma_{x_1} \sigma_{x_2}} = \frac{\int_{a_1}^{b_1} \int_{a_2}^{b_2} (x_1 - m_{x_1})(x_2 - m_{x_2}) p(x_1, x_2) dx_1 dx_2}{\sigma_{x_1} \sigma_{x_2}} \quad (19)$$

- If X_1 and X_2 are **independent**, then their joint PDF $p(x_1, x_2) = p_1(x_1) \cdot p_2(x_2)$.

$$\rho_{X_1, X_2} = \frac{\int_{a_1}^{b_1} (x_1 - m_{x_1}) p_1(x_1) dx_1}{\sigma_{x_1}} \cdot \frac{\int_{a_2}^{b_2} (x_2 - m_{x_2}) p_2(x_2) dx_2}{\sigma_{x_2}} = 0 \quad (20)$$

so $\text{Cov}(X_1, X_2) = 0$, the variance of X_3

$$\text{Var}[X_3] = c_1^2 \sigma_{x_1}^2 + c_2^2 \sigma_{x_2}^2 \quad (21)$$

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3. Sample Statistics

3.1. Sample Mean

3.2. Sample Variance

3.3. Precision and Accuracy

3.4. Summary of the important concepts

- The objective of any statistical method is to **estimate averages and associated variances and confidence levels** on the basis of a sampling process. We will look at the formulations of sample mean and sample variance.
- Based on the definition of the **true mean**, we define a formulation for the **sample mean, or sample average** of sample size N as follows:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \quad (22)$$

where $\{x_i\}_{i=1}^N$ are N **observations**, or **samples**, drawn from the RV X .

- When $N \rightarrow +\infty$, the sample mean needs to approach the true mean. This means that we have to examine the **expectation, or mean value of the sample mean**:

$$\begin{aligned}\mathbb{E}[\bar{x}] &= \mathbb{E}\left[\frac{1}{N} \sum_{i=1}^N x_i\right] \\ &= \frac{1}{N} \sum_{i=1}^N \mathbb{E}[x_i] \\ &= \frac{1}{N} \sum_{i=1}^N m_x \\ &= m_x\end{aligned}\tag{23}$$

- The sample mean $\bar{x}$ is an **unbiased estimator** of the **true mean**.

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3. Sample Statistics

3.1. Sample Mean

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- The **sample variance** is defined as:

$$s_x^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 \quad (24)$$

- Its **expectation** value:

$$\begin{aligned}
 \mathbb{E}[s_x^2] &= \mathbb{E}\left[\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2\right] \\
 &= \mathbb{E}\left[\frac{1}{N} \sum_{i=1}^N (x_i - m_x + m_x - \bar{x})^2\right] \\
 &= \frac{1}{N} \sum_{i=1}^N \mathbb{E}[(x_i - m_x)^2] + \frac{1}{N} \sum_{i=1}^N \mathbb{E}[(m_x - \bar{x})^2] + 2\mathbb{E}\left[(m_x - \bar{x}) \frac{1}{N} \sum_{i=1}^N (x_i - m_x)\right] \\
 &= \frac{1}{N} \sum_{i=1}^N \sigma_x^2 + \frac{1}{N} N \mathbb{E}[(m_x - \bar{x})^2] - 2\mathbb{E}[(m_x - \bar{x})^2] \\
 &= \sigma_x^2 - \mathbb{E}[(m_x - \bar{x})^2]
 \end{aligned}$$

where $\mathbb{E}[(m_x - \bar{x})^2] = \sigma_{\bar{x}}^2 = \text{Var}[\bar{x}]$ is the **variance of the sample mean $\bar{x}$** .

- The **variance of the sample mean** $\bar{x}$ is:

$$\begin{aligned}
 \text{Var}[\bar{x}] &= \sigma_{\bar{x}}^2 = \mathbb{E}[(m_x - \bar{x})^2] \\
 &= \text{Var}\left[\frac{1}{N} \sum_{i=1}^N x_i\right] \\
 &= \frac{1}{N^2} \sum_{i=1}^N \text{Var}[x_i] \\
 &= \frac{1}{N^2} \cdot N \cdot \sigma_x^2 = \frac{\sigma_x^2}{N}
 \end{aligned} \tag{25}$$

- The above formulation provides an important information that **the variance of the sample mean/average reduces with increasing number of samples**.
- Revisit the expectation value of the sample variance $\mathbb{E}[s_x^2]$:

$$\mathbb{E}[s_x^2] = \sigma_x^2 - \mathbb{E}[(m_x - \bar{x})^2] = \sigma_x^2 - \frac{\sigma_x^2}{N} = \frac{N-1}{N} \sigma_x^2 \tag{26}$$

- It shows that the sample variance s_x^2 is a **biased estimator** of the **true variance**.

- To obtain an unbiased formulation for the sample variance, we introduce:

$$\mathbb{E} \left[\frac{N}{N-1} s_x^2 \right] = \sigma_x^2 \quad (27)$$

- Hence, the **unbiased sample variance** S_x^2 is defined as:

$$\begin{aligned} S_x^2 &= \frac{N}{N-1} s_x^2 \\ &= \frac{N}{N-1} \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 \\ &= \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \end{aligned} \quad (28)$$

- In statistics, this formula is referred to as the **Bessel's correction**.

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3. Sample Statistics

3.1. Sample Mean

3.2. Sample Variance

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3.4. Summary of the important concepts

- To determine the confidence level on the outcome of a statistical process, it is necessary to determine the precision and accuracy associated with the **sample mean**.
- The **precision** is measured by the **relative statistical uncertainty** that is defined by:

$$R_x = \frac{\sigma_x}{\bar{x}} \quad (29)$$

Note that the **standard deviation** (σ_x) is also referred to as the **statistical uncertainty**.

- The **accuracy** refers to the measure of the deviation from the truth, i.e., how far is the (precise) sample mean from the true mean?
- Generally, to estimate the accuracy, it is necessary to either perform an experiment or to compare with the results of another formulation/technique that is known to be accurate.

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3. Sample Statistics

3.1. Sample Mean

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■ True mean:

$$m_x = \mathbb{E}[X] \quad (30)$$

■ True variance:

$$\text{Var}[X] = \sigma_x^2 = \mathbb{E}[(X - m_x)^2] \quad (31)$$

■ Sample mean:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \quad (32)$$

■ Expectation (mean value) of sample mean:

$$\mathbb{E}[\bar{x}] = m_x \quad (33)$$

■ Variance of sample mean:

$$\text{Var}[\bar{x}] = \sigma_{\bar{x}}^2 = \frac{\sigma_x^2}{N} \quad (34)$$

■ Sample variance (biased):

$$s_x^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 \quad (35)$$

■ Sample variance (unbiased):

$$S_x^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \quad (36)$$

■ Expectation of sample variance:

$$\begin{aligned} \mathbb{E}[s_x^2] &= \frac{N-1}{N} \sigma_x^2 \\ \mathbb{E}[S_x^2] &= \sigma_x^2 \end{aligned}$$

■ Variance of sample variance is not of interest in this class.

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4. Common Density Functions

4.1. Uniform Distribution

4.2. Bernoulli Distribution

4.3. Binomial Distribution

4.4. Geometric Distribution

4.5. Poisson Distribution

4.6. Normal/Gaussian Distribution

4.7. χ^2 Distribution

The uniform distribution is a continuous distribution

- If a RV X follows a uniform distribution over $[a, b]$, its PDF is a constant $p(x) = k$.
- The value of k can be obtained by normalization:

$$\int_a^b p(x)dx = 1, \quad p(x) = \frac{1}{b-a}, \quad a \leq x \leq b \quad (37)$$

- The **true mean** of X is:

$$\begin{aligned}\mathbb{E}[X] &= m_x = \int_a^b xp(x)dx \\ &= \int_a^b \frac{x}{b-a}dx \\ &= \frac{a+b}{2}\end{aligned}\tag{38}$$

- The **true variance** of X is:

$$\begin{aligned}\text{Var}[X] &= \sigma_x^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \int_a^b x^2 p(x)dx - \left(\frac{a+b}{2}\right)^2 \\ &= \frac{b^3 - a^3}{3(b-a)} - \left(\frac{a+b}{2}\right)^2 \\ &= \frac{(b-a)^2}{12}\end{aligned}\tag{39}$$

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4. Common Density Functions

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The Bernoulli distribution is a binary discrete distribution

- A **Bernoulli process** refers to a random process that has only **two outcomes** and the probabilities of these outcomes remain **constant** throughout the experimentation.
- Every RV in a Bernoulli process follows the identically distributed and independent (i.i.d.) **Bernoulli distribution**.

The Bernoulli distribution is a binary discrete distribution

- A **Bernoulli process** refers to a random process that has only **two outcomes** and the probabilities of these outcomes remain **constant** throughout the experimentation.
- Every RV in a Bernoulli process follows the identically distributed and independent (i.i.d.) **Bernoulli distribution**.
- A Bernoulli RV takes the value **1 (success)** with probability p , and the value **0 (failure)** with probability $q = 1 - p$. Its PMF is given by:

$$f(k; p) = p^k \cdot (1 - p)^{1-k}, \quad \text{for } k = 0, 1 \quad (40)$$

where k refers to the outcome. To avoid confusion with p , we will use f to denote the PDF/PMF in the following.

- The outcome of a coin toss follows a Bernoulli distribution. If the coin is fair, $p = 0.5$.

- Its **true mean** is:

$$\begin{aligned}
 \mathbb{E}[X] = m_x &= \sum_{k=0}^1 k f(k; p) \\
 &= \sum_{k=0}^1 k p^k \cdot (1-p)^{1-k} \\
 &= 0 + p = p
 \end{aligned} \tag{41}$$

- Its **true variance** is:

$$\begin{aligned}
 \text{Var}[X] = \sigma_x^2 &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\
 &= \sum_{k=0}^1 k^2 p^k \cdot (1-p)^{1-k} - p^2 \\
 &= p - p^2 \\
 &= p(1-p) \\
 &= pq
 \end{aligned} \tag{42}$$

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The Binomial distribution is a discrete distribution

- If a Bernoulli RV X is repeated N times, with outcomes as x_i 's, then the **sum of these outcomes** $Y = \sum_{i=1}^N x_i$ follows a **binomial distribution**.
- Given the probability of success p , **the probability of n successes out of N experiments** is:

$$\begin{aligned} P(Y = n) = f(n; N, p) &= \binom{N}{n} p^n q^{N-n} \\ &= \frac{N!}{(N-n)! n!} p^n q^{N-n}, \quad \text{for } n = 1, 2, \dots, N \end{aligned} \tag{43}$$

where $\binom{N}{n}$ is called the **binomial coefficient**.

- It refers to the number of combinations of N Bernoulli experiments with a sum of n , **irrespective of the order** of successful experiments.

- Its **true mean** is:

$$\mathbb{E}[Y] = m_y = \mathbb{E}\left[\sum_{i=1}^N x_i\right] = \sum_{i=1}^N \mathbb{E}[X] = Np \quad (44)$$

- Its **true variance** is:

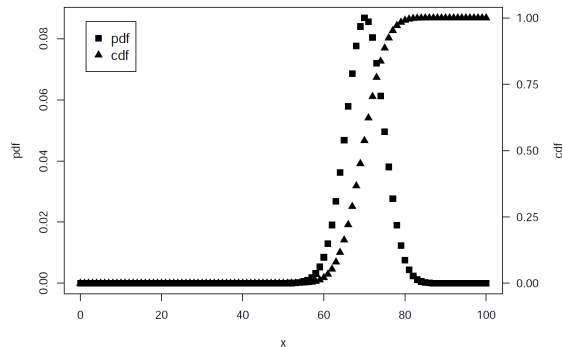
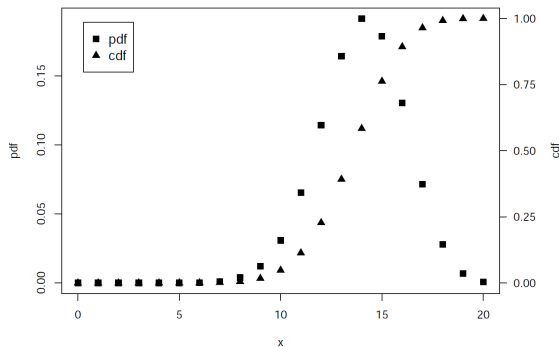
$$\text{Var}[Y] = \sigma_y^2 = \text{Var}\left[\sum_{i=1}^N x_i\right] = \sum_{i=1}^N \text{Var}[X] = Npq \quad (45)$$

- The **relative uncertainty** is given by:

$$R_y = \frac{\sigma_y}{m_y} = \frac{\sqrt{Npq}}{Np} = \sqrt{\frac{q}{Np}} \quad (46)$$

which indicates that the relative uncertainty decreases with increasing number of experiments.

- Left: $p = 0.7, N = 20$, right: $p = 0.7, N = 100$
- If N is large enough, the skewness (a measure of the asymmetry of the probability distribution) of the distribution is not too great.
- In this case a **binomial distribution approximates a normal distribution** $\mathcal{N}(Np, Np(1-p))$. Empirically, such normal approximation can be used when $Np \geq 5$ and $N(1-p) \geq 5$.



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A geometric distribution is a discrete distribution

- A geometric distribution gives the probability of achieving a success following $(n - 1)$ failures. Sometimes it is also defined as the number of trials to achieve a first success.
- A geometric RV X has PMF:

$$P(X = n) = f(n; p) = (1 - p)^{n-1}p = q^{n-1}p, \quad \text{for } n = 1, 2, \dots, N \quad (47)$$

where p is the probability of success, q is the probability of failure, and n is the number of trials.

■ The **true mean** of X is:

$$\begin{aligned}\mathbb{E}[X] = m_x &= \sum_{n=1}^{\infty} nq^{n-1}p \\ &= p(1 + 2q + 3q^2 + 4q^3 + \dots) \\ &= p \left[\sum_{n=1}^{\infty} q^{n-1} + \sum_{n=2}^{\infty} q^{n-1} + \sum_{n=3}^{\infty} q^{n-1} + \dots \right] \\ &= p \left[\frac{1}{1-q} + \frac{q}{1-q} + \frac{q^2}{1-q} + \dots \right] \\ &= 1 + q + q^2 + \dots \\ &= \frac{1}{1-q} \\ &= \frac{1}{p}\end{aligned}\tag{48}$$

- The **true variance** of X is:

$$\begin{aligned}
 \text{Var}[X] &= \sigma_x^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\
 &= \sum_{n=1}^{\infty} n^2 q^{n-1} p - \frac{1}{p^2} \\
 &= \sum_{n=1}^{\infty} n(n+1) q^{n-1} p - \sum_{n=1}^{\infty} n q^{n-1} p - \frac{1}{p^2} \\
 &= \sum_{n=1}^{\infty} n(n+1) q^{n-1} p - \frac{1}{p} - \frac{1}{p^2} \\
 &= p \frac{\partial^2}{\partial q^2} \sum_{n=1}^{\infty} q^{n+1} - \frac{1}{p} - \frac{1}{p^2} \\
 &= p \frac{\partial^2}{\partial q^2} \left[\frac{1}{1-q} - 1 - q \right] - \frac{1}{p} - \frac{1}{p^2} \\
 &= \frac{2}{p^2} - \frac{1}{p} - \frac{1}{p^2} = \frac{1}{p^2} - \frac{1}{p} = \frac{q}{p^2}
 \end{aligned}$$

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The Poisson distribution is a discrete distribution

- The **binomial distribution approaches a Poisson distribution** if we have the following conditions:

$$p \ll 1, \quad N \gg 1, \quad n \ll N \quad (49)$$

- Given the binomial distribution as:

$$f(n; N, p) = \binom{N}{n} p^n q^{N-n} = \frac{N \cdot (N-1) \cdot (N-2) \dots (N-n+1)}{n!} \cdot p^n \cdot (1-p)^{N-n} \quad (50)$$

- Since the true mean of a binomial distribution $m = Np$, we have $p = \frac{m}{N}$.

- Also since $n \ll N$:

$$f(n; N, p) \cong \frac{N^n}{n!} \left(\frac{m}{N}\right)^n \left(1 - \frac{m}{N}\right)^{N-n} \cong \frac{m^n}{n!} \left(1 - \frac{m}{N}\right)^{-n} \left(1 - \frac{m}{N}\right)^N \quad (51)$$

- For the second term on the right hand side

$$\lim_{N \rightarrow \infty} \left(1 - \frac{m}{N}\right)^{-n} \approx 1 \quad (52)$$

- For the third term on the right-hand side:

$$\lim_{N \rightarrow \infty} \left(1 - \frac{m}{N}\right)^N \quad (53)$$

- We need to use the definition of e , which is

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x \quad (54)$$

- We can define

$$x = -\frac{N}{m} \quad (55)$$

- In this way

$$\lim_{N \rightarrow \infty} \left(1 - \frac{m}{N}\right)^N = \lim_{N \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{x(-m)} = e^{-m} \quad (56)$$

- Finally, the PMF becomes:

$$f(n; N, p) = \frac{m^n}{n!} e^{-m} \quad (57)$$

- Define X as a Poisson RV, it is common in literature to write its PMF as:

$$f(k; \lambda) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad \text{for } k = 0, 1, 2, \dots \quad (58)$$

where k is the number of occurrences, λ is called the **rate constant**.

- The Poisson distribution expresses **the probability of a given number of events occurring in a fixed interval of time or space if these events occur with a known constant mean rate and independently of the time since the last event.**

- The **true mean** of X :

$$\begin{aligned}
 \mathbb{E}[X] = m_x &= \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} e^{-\lambda} \\
 &= \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \\
 &= \lambda e^{-\lambda} \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} \\
 &= \lambda e^{-\lambda} \frac{1}{e^{-\lambda}} = \lambda
 \end{aligned} \tag{59}$$

- Note that we have used

$$\sum_{n=0}^{\infty} \frac{\lambda^n}{n!} = \frac{1}{e^{-\lambda}} \tag{60}$$

because

$$\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} = 1 \tag{61}$$

- The **true variance** of X :

$$\begin{aligned}
 \text{Var}[X] &= \sigma_x^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \mathbb{E}[X^2] - \lambda^2 \\
 &= \sum_{k=0}^{\infty} k^2 \frac{\lambda^k}{k!} e^{-\lambda} - \lambda^2 \\
 &= \lambda \sum_{k=1}^{\infty} k \frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda} - \lambda^2 \\
 &= \lambda \sum_{n=0}^{\infty} (n+1) \frac{\lambda^n}{n!} e^{-\lambda} - \lambda^2 \\
 &= \lambda \left[\sum_{n=0}^{\infty} n \frac{\lambda^n}{n!} e^{-\lambda} + \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} e^{-\lambda} \right] - \lambda^2 \\
 &= \lambda [\lambda + 1] - \lambda^2 = \lambda
 \end{aligned}$$

- Hence, the **relative uncertainty** is given by:

$$R_x = \frac{\sigma_x}{m_x} = \frac{\sqrt{\lambda}}{\lambda} = \frac{1}{\sqrt{\lambda}} \quad (62)$$

Outline

4. Common Density Functions

4.1. Uniform Distribution

4.2. Bernoulli Distribution

4.3. Binomial Distribution

4.4. Geometric Distribution

4.5. Poisson Distribution

4.6. Normal/Gaussian Distribution

4.7. χ^2 Distribution

The Gaussian distribution is a continuous distribution

- For a large number of experiments (N), the **de Moivre-Laplace limit theorem** states that a binomial distribution function approaches a normal (Gaussian) distribution function.
- Recall the **binomial distribution function (PMF)**:

$$f(n; N, p) = \binom{N}{n} p^n q^{N-n} = \frac{N!}{(N-n)! n!} p^n q^{N-n}, \quad \text{for } n = 1, 2, \dots, N \quad (63)$$

- The de Moivre-Laplace theorem states that, as N grows large, for n in the neighborhood of Np we can approximate

$$\binom{N}{n} p^n q^{N-n} \cong \frac{1}{\sqrt{2\pi Npq}} \cdot e^{-\frac{(n-Np)^2}{2Npq}}, \quad p+q=1, \quad p, q > 0 \quad (64)$$

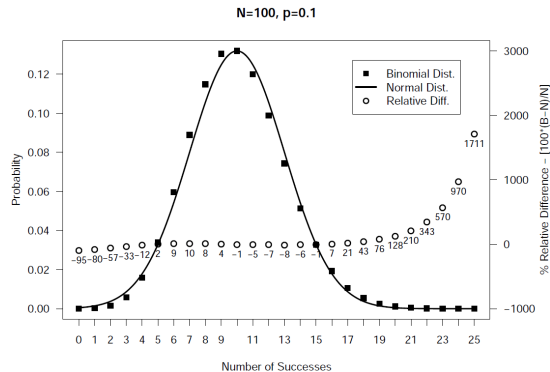
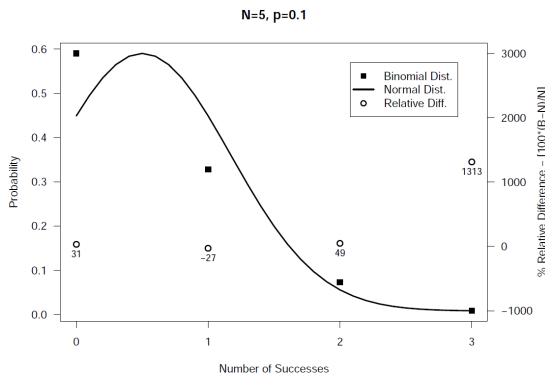
in the sense that the ratio of the left-hand side to the right-hand side converges to 1 as $N \rightarrow \infty$.

- The derivation can be found in the textbook. It will not be covered in the lecture.

- Previously we have derived that, for a binomial PDF, the true mean and variance are given by $m = Np$ and $\sigma^2 = Npq$, respectively. Using these formulations, we obtain the formulation of the normal PDF for random variable X as follows:

$$f(x; m, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-m)^2}{2\sigma^2}}, \quad -\infty < x < \infty \quad (65)$$

- To examine the accuracy of a normal density function for approximating a binomial density function, we consider a Bernoulli process with $p = 0.1$.



- The normal/Gaussian distribution has **two parameters m and σ^2** , which are its **true mean** and **true variance**.
- A univariate Gaussian distribution is typically denoted as $\mathcal{N}(m, \sigma^2)$.
- If $X \sim \mathcal{N}(m, \sigma^2)$, then the RV $Z = \frac{X-m}{\sigma}$ follows a $\mathcal{N}(0, 1)$ distribution, which is called the **standard normal/Gaussian** distribution.

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, \quad -\infty < z < \infty \quad (66)$$

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$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, \quad -\infty < z < \infty \quad (66)$$

- The CDF of the standard normal $Z \sim \mathcal{N}(0, 1)$ is defined as:

$$\Psi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt = \left(\int_{-\infty}^0 + \int_0^z \right) \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{z}{\sqrt{2}} \right) \right] \quad (67)$$

where the **error function $\operatorname{erf}(x)$** is defined as:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad (68)$$

■ **Statistical moments** for $X \sim \mathcal{N}(m, \sigma^2)$:

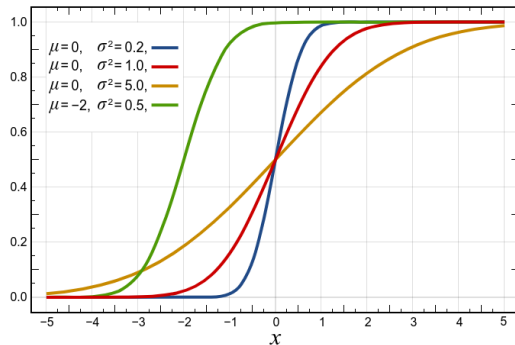
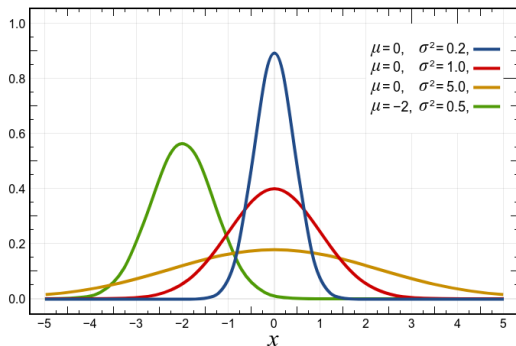
$$\mathbb{E}[X] = m$$

$$\mathbb{E}[X - m] = 0$$

$$\mathbb{E}[X^2] = (\mathbb{E}[X])^2 + \text{Var}[X] = m^2 + \sigma^2$$

$$\mathbb{E}[(X - m)^2] = \text{Var}[X] = \sigma^2$$

■ Left: PDFs, right: CDFs. The normal PDF peaks at m , and it is symmetric about the mean.



The 68-95-99.7 rule

- A normal distribution is sometimes informally called a **bell curve**.
- However, note that many other distributions are bell-shaped (such as the Cauchy, Student's t , and logistic distributions).

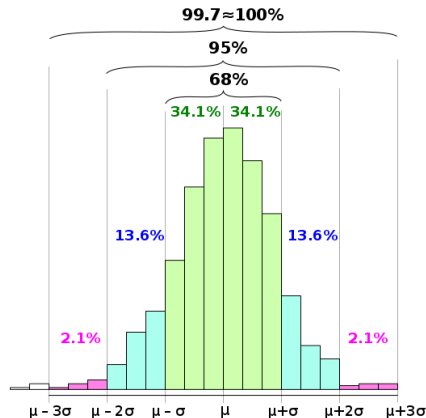
- With $X \sim \mathcal{N}(m, \sigma^2)$, and $Z \sim \mathcal{N}(0, 1)$,
 $X = \sigma Z + m$:

$$P(|X - m| \leq \sigma) = P(|Z| \leq 1) = 0.6826$$

$$P(|X - m| \leq 2\sigma) = P(|Z| \leq 2) = 0.9544$$

$$P(|X - m| \leq 3\sigma) = P(|Z| \leq 3) = 0.9974$$

- In statistics, the **68-95-99.7 rule**, also known as the **empirical rule**, or the **three-sigma rule of thumb**:, 68.27%, 95.45% and 99.73% of the values lie within **one**, **two** and **three** standard deviations of the mean, respectively.



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4.7. χ^2 Distribution

- The chi-square distribution (also chi-squared or χ^2 -distribution) with k degrees of freedom is the distribution of a sum of the squares of k **independent standard normal** RVs.
- The chi-square distribution is a special case of the **gamma distribution**.
- If $Z_1, Z_2, \dots, Z_k$ are **independent, standard normal RVs**, then the sum of their squares is distributed according to the χ^2 -distribution:

$$\sum_{i=1}^k Z_i^2 \sim \chi^2(k), \text{ or } \chi_k^2 \quad (69)$$

where the positive integer k is called the **degrees of freedom**.

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- PDF:

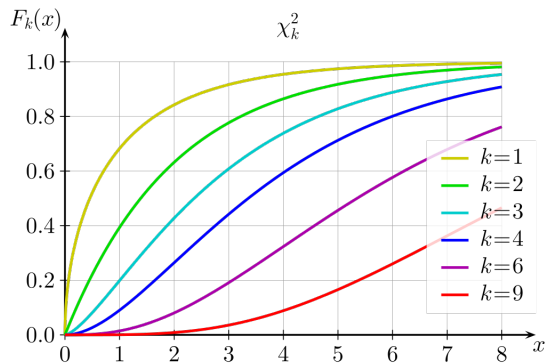
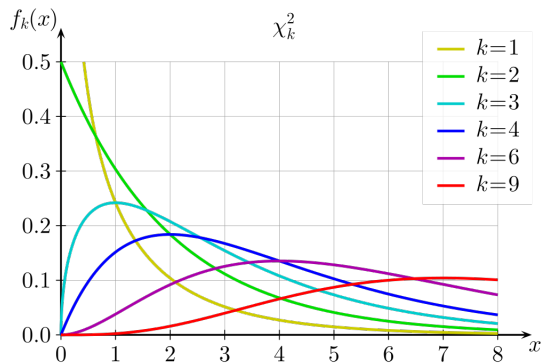
$$f(x; k) = \frac{x^{\frac{k}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{k}{2}} \Gamma\left(\frac{k}{2}\right)}, \quad \text{for } 0 < x < \infty \quad (70)$$

where Γ is the gamma function.

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt \quad (71)$$

- $\chi^2(k)$ has a mean value of k and variance of $2k$.

- Left: PDF of χ_k^2 with different degrees of freedom.
- Right: CDF of χ_k^2 with different degrees of freedom.



Outline

1. Introduction
2. Expectation
3. Sample Statistics
4. Common Density Functions
- 5. Limit Theorems**
6. Relative Uncertainty
7. Confidence Levels
8. Normality Tests

Outline

5. Limit Theorems

5.1. The de Moivre-Laplace Limit Theorem

5.2. The Central Limit Theorem

- The **de Moivre-Laplace limit theorem** and the **central limit theorem** are commonly used for **estimating the confidence level** on the results of a statistical analysis.
- Suppose we want to determine the probability that the **sampled probability of success (p')** of a **Bernoulli process** is within a certain range (ϵ) of the **true probability of success (p)** as follows:

$$P(|p' - p| \leq \epsilon) = ? \quad (72)$$

- Denote the number of successes in N trials as x , we have $p' = x/N$. The inequality becomes:

$$\left| \frac{x}{N} - p \right| \leq \epsilon \quad (73)$$

- Multiple both sides by N , and then divide both sides by $\sqrt{Npq}$:

$$\left| \frac{x - Np}{\sqrt{Npq}} \right| \leq \epsilon \sqrt{\frac{N}{pq}} \quad (74)$$

- The **de Moivre-Laplace theorem** states that at the limit of large number of experiments (N), the **binomial distribution** is represented by a **normal distribution** for the RV X with true mean of $m_x = Np$, and true variance of $\sigma_x^2 = Npq$.
- Given $m_x = Np$ and $\sigma_x^2 = Npq$:

$$\left| \frac{x - m_x}{\sigma_x} \right| \leq \epsilon \sqrt{\frac{N}{pq}} \quad (75)$$

- Define a **new RV** $T = \frac{X - m_x}{\sigma_x}$, which follows $\mathcal{N}(0, 1)$. The inequality becomes:

$$|t| \leq \epsilon \sqrt{\frac{N}{pq}} \quad (76)$$

- The probability that the above inequality is true can be obtained by using the CDF for T :

$$P \left(|t| \leq \epsilon \sqrt{\frac{N}{pq}} \right) = 2\Psi \left(\epsilon \sqrt{\frac{N}{pq}} \right) - 1 \quad (77)$$

- Hence, the **corollary to the de Moivre-Laplace limit theorem** is expressed as:

$$P(|p' - p| \leq \epsilon) = 2\Psi \left(\epsilon \sqrt{\frac{N}{pq}} \right) - 1 \quad (78)$$

Example 1

- For a Bernoulli RV, after $N = 100$ experiments, $x = 8$ successes are achieved. What is the level of confidence that the sampled probability is within $\epsilon = 1.0\%$ of the true probability?
- The sampled probability is $p' = \frac{x}{N} = \frac{8}{100} = 0.08$.
- The confidence can be estimated as:

$$\begin{aligned}
 P(|p' - p| \leq \epsilon) &= P(|0.08 - p| \leq 0.01) \\
 &= 2\Psi\left(\epsilon\sqrt{\frac{N}{pq}}\right) - 1 \\
 &= 2\Psi\left(0.01 \times \sqrt{\frac{100}{0.08 \times 0.92}}\right) - 1 \\
 &= 2\Psi(0.3686) - 1 \\
 &= 2 \times 0.644 - 1 \\
 &= 0.2876
 \end{aligned}$$

- This means that the confidence level for this experiment is only 28.76%.

Example 2

- How many experiments (N) are needed in order to achieve a 95% confidence on a ϵ of 0.2% for a Bernoulli RV?

$$P(|p' - p| \leq \epsilon) = 2\Psi\left(\epsilon\sqrt{\frac{N}{pq}}\right) - 1 = 0.95 \quad (79)$$

$$\Psi\left(\epsilon\sqrt{\frac{N}{pq}}\right) = 0.975 \quad (80)$$

- Using tables of the normal CDF, $\epsilon\sqrt{\frac{N}{pq}} = 1.96$. Hence, N is given by:

$$N = \left(\frac{1.96}{\epsilon}\right)^2 pq \approx 10^6 pq \quad (81)$$

- To be conservative, we consider the maximum value of pq that corresponds to $p = q = \frac{1}{2}$. Hence $N = 250,000$.

Discussions of Example 2

- Assuming a maximum value for pq results in a large number of experiments (250,000) that may require a significant amount of computer time.
- Therefore, it is recommended to conduct initial sampling to estimate p' , and therefore avoid unnecessary waste of resources.
- If the estimated p' is 0.01, then the corollary predicts a reduced number of experiments of only about 10,000 that is smaller by a factor of 25!
- This is true for the corollary that has been derived based on the **absolute difference** of the sampled and true probabilities of success.

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- This is true for the corollary that has been derived based on the **absolute difference** of the sampled and true probabilities of success.
- To examine the **precision** of a sample mean, however, it is necessary to examine its **relative uncertainty** defined as:

$$\frac{\sigma_x}{\bar{x}} \approx \frac{\sigma_x}{m_x} = \frac{\sqrt{Npq}}{Np} = \sqrt{\frac{q}{Np}} \quad (82)$$

- **These two cases have the same 95% confidence, but their precisions differ by a factor of 50.**
 - **Case 1:** $N = 250,000$, $p = q = 0.5$ has a relative uncertainty of $\frac{1}{500} = 0.2\%$, which is a very precise estimation.
 - **Case 2:** $N = 10,000$, $p = 0.01$, $q = 0.99$ has a relative uncertainty of about 10%, which is a very imprecise estimation.

- To avoid such observation, we can derive the formulation for the corollary based on the **relative difference** $\left| \frac{p' - p}{p} \right|$, instead of the **absolute difference** $|p' - p|$, of the sample mean and true mean as follows:

$$\left| \frac{p' - p}{p} \right| \leq \epsilon \quad (83)$$

- Similarly, multiply both sides by $\sqrt{Np/q}$:

$$\left| \frac{x - m_x}{\sigma_x} \right| \leq \epsilon \sqrt{\frac{Np}{q}} \quad (84)$$

- Hence, the probability for the relative difference of the sampled and true success probabilities is expressed by:

$$P \left(\left| \frac{p' - p}{p} \right| \leq \epsilon \right) = 2\Psi \left(\epsilon \sqrt{\frac{Np}{q}} \right) - 1 \quad (85)$$

Example 3

- If we use the corollary based on the **relative difference** for Example 2, the predicted number of experiments is given by:

$$N \approx 10^6 \cdot \left(\frac{q}{p} \right) \quad (86)$$

- For $p' = 0.5$, $N \approx 10^6 \times \frac{0.5}{0.5} = 10^6$.
- For $p' \approx 0.01$, $N \approx 10^6 \times \frac{0.99}{0.01} = 10^8$.
- The above predicted number of experiments is more realistic, because intuitively more experiments are needed when the probability of success is smaller.
- Now in case 2, $N = 10^8$, $p = 0.01$, $q = 0.99$ has a relative uncertainty of about 0.1%, which is also a very precise estimation.

Outline

5. Limit Theorems

5.1. The de Moivre-Laplace Limit Theorem

5.2. The Central Limit Theorem

The Central Limit Theorem (CLT)

- The CLT establishes that, when independent RVs are added, their properly normalized sum tends toward a normal distribution even if the original RVs themselves are not normally distributed.
- Let $\{x_1, x_2, \dots, x_N\}$ be a sequence of **independent** samples from an **arbitrary** RV X , whose mean and variance are finite:

$$\mathbb{E}[X] = m_x, \quad \text{Var}[X] = \sigma_x^2 < +\infty \quad (87)$$

- Define the sample mean

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \quad (88)$$

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$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \quad (88)$$

- Let $G_N(x)$ denote the CDF of $\frac{(\bar{x}-m_x)}{\sigma_x/\sqrt{N}}$. Then:

$$\lim_{N \rightarrow \infty} G_N(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy, \quad -\infty < x < \infty \quad (89)$$

that is, $\frac{(\bar{x}-m_x)}{\sigma_x/\sqrt{N}}$ has a **limiting standard normal distribution**, even if the original RVs are **NOT** normally distributed.

- The CLT theorem implies that probabilistic and statistical methods that work for normal distributions can be applicable to many problems involving other types of distributions.

- The limiting PDF of the sample mean $\bar{x}$ is:

$$f(\bar{x}) = \frac{1}{\sqrt{2\pi\sigma_{\bar{x}}^2}} e^{-\frac{(\bar{x}-m_x)^2}{2\sigma_{\bar{x}}^2}}, \quad -\infty < \bar{x} < \infty \quad (90)$$

where $\sigma_{\bar{x}}^2 = \frac{\sigma_x^2}{N}$.

- Provided that N is sufficiently large, we may use the CLT to estimate a reduced variance for $\bar{x}$. When σ_x^2 is unknown, we can approximate it by calculating the (unbiased) sample variance S_x^2 .

$$\sigma_x^2 \approx S_x^2, \quad \sigma_{\bar{x}}^2 \approx \frac{S_x^2}{N} \quad (91)$$

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$$\sigma_x^2 \approx S_x^2, \quad \sigma_{\bar{x}}^2 \approx \frac{S_x^2}{N} \quad (91)$$

- We may estimate the confidence level of the true mean based on $\sigma_{\bar{x}}$ as follows:
 - Considering a change of variable $Z = \frac{\bar{x}-m_x}{\sigma_{\bar{x}}}$.
 - The probability that Z is within n standard deviations ($\sigma_{\bar{x}}$) is given by:

$$P[-n \leq Z \leq n] = 2\Psi(n) - 1 \quad (92)$$

with confidence levels of 68.3%, 95.4%, and 99.7% for n equals to 1, 2, and 3, respectively.

- Clearly, the de Moivre-Laplace limit theorem is a special case of the CLT. The former is defined only for Bernoulli RVs, while CLT is defined for any type of RVs.

Demonstration of the CLT

- Consider a MC simulation consisting of T trials, where each trial has N experiments. What are the sample mean and the sample variance?
- The sample mean in the k^{th} trial is:

$$\bar{x}_k = \frac{1}{N} \sum_{i=1}^N x_{ki}, \quad \text{for } k = 1, 2, \dots, T \quad (93)$$

- Define the sample mean for all trials as:

$$\langle \bar{x} \rangle = \frac{1}{T} \sum_{k=1}^T \bar{x}_k \quad (94)$$

- The sample variance for all trials is obtained by:

$$S_x^2 = \frac{1}{T-1} \sum_{k=1}^T (\bar{x}_k - \langle \bar{x} \rangle)^2 \quad (95)$$

- Without knowing CLT, the sample mean $\langle \bar{x} \rangle$ is an **unbiased** estimator of the true mean m_x , we will assume it has a sample variance S_x^2 .
- If we further assume that $\langle \bar{x} \rangle$ follows a normal distribution, then we may state that the true mean m_x is within

$$|\langle \bar{x} \rangle - m_x| \leq n \cdot S_x \quad (96)$$

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with confidence levels of 68.3%, 95.4%, and 99.7% for n equals to 1, 2, and 3, respectively.

- However, CLT provides a **much stronger claim**. With the CLT, $\langle \bar{x} \rangle$ is **normally distributed** about the true mean m_x with a **variance of $\frac{S_x^2}{T}$** , which is smaller than S_x^2 . This claim is true no matter what distribution the RV X follows.
- Even though we have calculated only one value of $\langle \bar{x} \rangle$, we may state that the true mean m_x is within

$$|\langle \bar{x} \rangle - m_x| \leq n \cdot \frac{S_x}{\sqrt{T}} \quad (97)$$

with confidence levels of 68.3%, 95.4%, and 99.7% for n equals to 1, 2, and 3, respectively.

- Consider a RV X , $p(x) = 2e^{-2x}$, $0 \leq x < +\infty$, with $m_x = 0.5$ and $\sigma_x^2 = 0.25$. The corresponding FPMC is given by

$$x = -\frac{\ln \eta}{2} \quad (98)$$

Then we can use a computer code to generate 1000 samples of X .

- Then we estimate the sample mean and sample variance for different grouping of the 1000 experiments into **Experiments per Trial (N)** and **Trials T** as presented in Table 4.4.
- Table 4.4 shows that the sample mean is the same for all the combinations of N and T , and it is close to the true mean of 0.5.
- The sample variance S_x^2 decreases with increasing N . It is approximately proportional to $\frac{1}{N}$ as is expected by the CLT.

Table 4.4 Demonstration of the Central Limit Theorem

Experiments Trial(N)	Trials (T)	Sample mean ($\bar{x}$)	sample variance (S_x^2)
1	1000	0.498	0.249
2	500	0.498	0.119
4	250	0.498	0.060
8	125	0.498	0.0307
10	100	0.498	0.0252
40	25	0.498	0.00958
100	10	0.498	0.00344
200	5	0.498	0.00078
500	2	0.498	0.000061

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1. Introduction
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- 6. Relative Uncertainty**
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- Previously, we have derived that, for a RV X with samples $\{x_1, x_2, \dots, x_N\}$, the unbiased sample variance estimator S_x^2 is defined as:

$$S_x^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \quad (99)$$

■ Expand the relation:

$$\begin{aligned}
 S_x^2 &= \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \\
 &= \frac{1}{N-1} \sum_{i=1}^N (x_i^2 + (\bar{x})^2 - 2x_i\bar{x}) \\
 &= \frac{1}{N-1} \sum_{i=1}^N x_i^2 + \frac{1}{N-1} \sum_{i=1}^N (\bar{x})^2 - \frac{2}{N-1} \sum_{i=1}^N x_i\bar{x} \\
 &= \frac{N}{N-1} \frac{1}{N} \sum_{i=1}^N x_i^2 + \frac{N}{N-1} (\bar{x})^2 - \frac{2N}{N-1} \bar{x} \frac{1}{N} \sum_{i=1}^N x_i \\
 &= \frac{N}{N-1} \overline{x^2} + \frac{N}{N-1} (\bar{x})^2 - \frac{2N}{N-1} (\bar{x})^2 \\
 &= \frac{N}{N-1} \overline{x^2} - \frac{N}{N-1} (\bar{x})^2 \\
 &= \frac{N}{N-1} \left(\overline{x^2} - (\bar{x})^2 \right)
 \end{aligned}$$

- Then, the relative uncertainty is given by:

$$R_x = \frac{\sqrt{S_x^2}}{\bar{x}} = \frac{\sqrt{\frac{N}{N-1} (\overline{x^2} - (\bar{x})^2)}}{\bar{x}} = \sqrt{\frac{N}{N-1} \left[\frac{\overline{x^2}}{(\bar{x})^2} - 1 \right]} \quad (100)$$

- According to the CLT, the relative uncertainty for the sample mean $\bar{x}$ is:

$$\begin{aligned}
 R_{\bar{x}} &= \frac{\sigma_{\bar{x}}}{\bar{x}} = \frac{\sigma_x / \sqrt{N}}{\bar{x}} \\
 &= \frac{R_x}{\sqrt{N}} \\
 &= \sqrt{\frac{1}{N-1} \left[\frac{\overline{x^2}}{(\bar{x})^2} - 1 \right]} \\
 &= \sqrt{\frac{\frac{1}{N} \sum_{i=1}^N x_i^2}{(N-1) \left[\frac{1}{N} \sum_{i=1}^N x_i \right]^2} - \frac{1}{N-1}} \\
 &= \sqrt{\frac{\sum_{i=1}^N x_i^2}{\left[\sum_{i=1}^N x_i \right]^2} - \frac{1}{N}}
 \end{aligned}$$

where the last equation was simplified for cases when $N \gg 1$, $\frac{1}{N-1} \approx \frac{1}{N}$.

Verify the relative uncertainty for a Bernoulli process

- Consider a Bernoulli process in which the outcomes are either 1 or 0.
- Assume that in N trials we have observed c successes, we have

$$\sum_{i=1}^N x_i^2 = c$$

and

$$\sum_{i=1}^N x_i = c$$

- The formulation for the relative uncertainty for a Bernoulli RV reduces to

$$R_{\bar{x}} = \sqrt{\frac{\sum_{i=1}^N x_i^2}{\left[\sum_{i=1}^N x_i\right]^2} - \frac{1}{N}} = \sqrt{\frac{c}{c^2} - \frac{1}{N}} = \sqrt{\frac{1}{c} - \frac{1}{N}} \quad (101)$$

- To verify this relation, we can derive $R_{\bar{x}}$ in a different way.
- The probability of success is $p' = c/N$. Knowing that the true mean and variance of a Bernoulli process are p and pq respectively, then we may derive the sample relative uncertainty as:

$$\begin{aligned}
 R_x &= \frac{S_x}{p'} \\
 &= \frac{\sqrt{\frac{c}{N} \left(1 - \frac{c}{N}\right)}}{\frac{c}{N}} \\
 &= \sqrt{\frac{N}{c} - 1}
 \end{aligned} \tag{102}$$

- Given the CLT, we have:

$$R_{\bar{x}} = \frac{R_x}{\sqrt{N}} = \sqrt{\frac{1}{c} - \frac{1}{N}} \tag{103}$$

which is the same with Equation (101).

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1. Introduction
2. Expectation
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Outline

7. Confidence Levels

7.1. Student's t -distribution

7.2. Determination of confidence level

Student's t -distribution: inference with small samples with unknown population variance

- We will discuss a method for estimating the confidence level in case of finite sampling.
- We will introduce the Student's t -distribution and then discuss its application for the estimation of the confidence level.
- Previously we learnt in the CLT that when the number of experiments N per trial is infinite, the deviation of the average from the true mean, i.e., $\frac{\bar{x} - m_x}{\sigma_{\bar{x}}}$ follows a standard normal distribution.
- This behavior, however, may NOT happen for a finite value of N .

Student's t -distribution: inference with **small samples with unknown population variance**

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- Previously we learnt in the CLT that when the number of experiments N per trial is **infinite**, the deviation of the average from the true mean, i.e., $\frac{\bar{x} - m_x}{\sigma_{\bar{x}}}$ follows a standard normal distribution.
- This behavior, however, may NOT happen for a **finite** value of N .
- The **Student's t -distribution** was developed by William Gossett to estimate a “better” variance for cases in which N is **finite**.
- Gossett observed that for relatively small N , generally a normal distribution **overestimates the values near the mean** and **underestimates them away from the mean**.
- Gossett derived a distribution considering:
 - 1 fitting the results of a sampling of finite N
 - 2 including an adjustable parameter or a degree of freedom

- Let $\{x_1, x_2, \dots, x_N\}$ be independently and identically drawn from a RV X with **true mean** m_x and **true variance** σ_x^2 .
- The **sample mean** and **(unbiased) sample variance estimators** are:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \quad \text{and} \quad S_x^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \quad (104)$$

- The CLT tells us that the RV

$$Z = \frac{\bar{x} - m_x}{\sigma_x / \sqrt{N}} \quad (105)$$

follows a standard normal distribution when $N \rightarrow \infty$.

- A new RV

$$T = \frac{\bar{x} - m_x}{S_x / \sqrt{N}} \quad (106)$$

follows a **Student's t -distribution** with $(N-1)$ degrees of freedom.

- Since S_x has replaced σ_x , the only **unobservable** quantity in this expression is m_x , so this can be used to derive confidence intervals for m_x .

- The PDF of Student's t -distribution is given by:

$$f(t; \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi} \cdot \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}} = \frac{1}{\sqrt{\nu} \cdot B\left(\frac{1}{2}, \frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}, -\infty < x < +\infty \quad (107)$$

where ν is the degree of freedom, Γ is the gamma function, and B is the Beta function. Sometimes the PDF is also written as $f_\nu(t)$.

- The true mean is

$$\mathbb{E}[T] = m_t = \int_{-\infty}^{\infty} f(t; \nu) t dt = 0, \quad \text{when } \nu > 1 \quad (108)$$

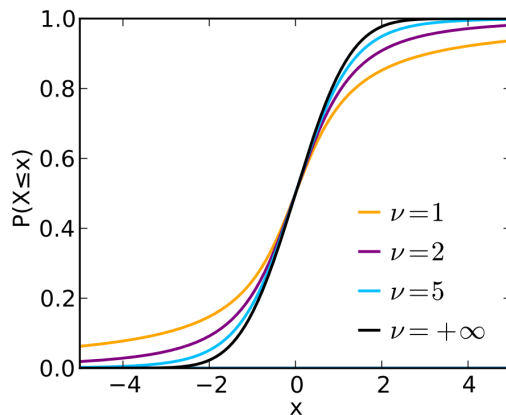
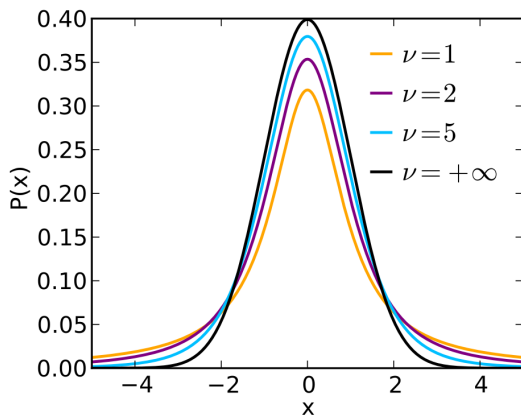
- The higher moments of T exist when the moment number (n) is less than the degree of freedom (ν). The odd moments are equal to zero because of symmetry, and the even moments are

$$\mathbb{E}[T^n] = \frac{\nu^{\frac{n}{2}} \cdot \Gamma\left(\frac{n+1}{2}\right) \cdot \Gamma\left(\frac{\nu-n}{2}\right)}{\sqrt{\pi} \cdot \Gamma\left(\frac{\nu}{2}\right)}, \quad \text{for } n < \nu \text{ and } n \text{ is even} \quad (109)$$

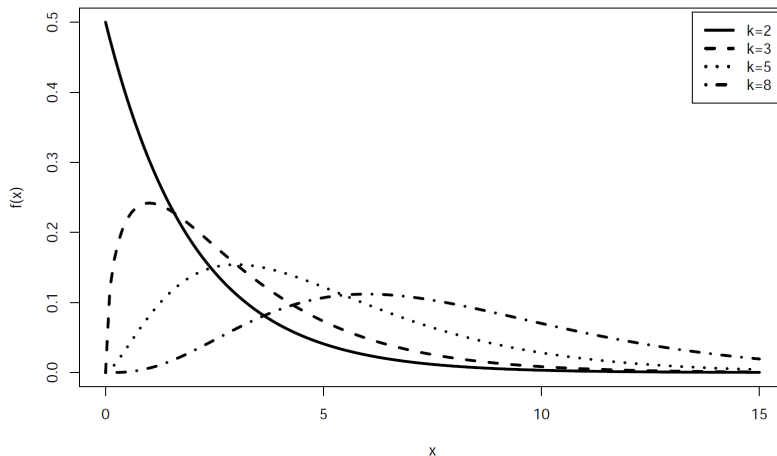
- The true variance is:

$$\sigma_t^2 = \mathbb{E}[T^2] - (\mathbb{E}[T])^2 = \frac{\nu}{\nu - 2}, \quad \text{for } \nu > 2 \quad (110)$$

- As the number of degrees of freedom ν goes to ∞ , the t -distribution approaches the standard normal distribution. Note that in practical applications, when $\nu > 30$, the t -distribution can be considered as a standard normal distribution.
- Left: PDF of Student's t -distribution with different ν . The figure indicates that the t -distribution has a lower value for the mean and higher values at the tails of the distribution.
- Right: CDF of Student's t -distribution with different ν .



- Recall the χ^2 -distribution. It has a lower mean and relatively higher tails as compared to a normal density function.
- One can sample a t -distribution using $t = \frac{z}{\sqrt{\chi^2/k}}$, where z and χ^2 are independent samples. z is sampled from a standard normal distribution, while χ^2 is sampled from a χ^2 -distribution with k degrees of freedom.



Outline

7. Confidence Levels

7.1. Student's t -distribution

7.2. Determination of confidence level

- A **confidence interval** gives an estimated **range** of values which is likely to include an unknown population parameter, the estimated range being calculated from **a given set of sample data**.
- The selection of a **confidence level** for an interval determines the **probability** that the confidence interval produced will **contain** the true parameter value.
- In the case of a **finite sample size**, there is a possibility that the distribution of the sample mean is not close enough to a normal distribution. Therefore, we employ the **Student's t -distribution** to achieve a higher confidence.

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- In the case of a **finite sample size**, there is a possibility that the distribution of the sample mean is not close enough to a normal distribution. Therefore, we employ the **Student's t -distribution** to achieve a higher confidence.
- Based on a t -distribution with $(N - 1)$ degrees of freedom, the **t -factor t_{N-1}** , which is equal to **the number of standard deviations** for achieving a **95% confidence level**, is obtained.
- The t -factor is used to determine the 95% confidence level d_{95} as

$$d_{95} = t_{N-1} \cdot S_{\bar{x}} = t_{N-1} \cdot \frac{S_x}{\sqrt{N}} \quad (111)$$

where $S_{\bar{x}}$ is the estimated sample standard deviation for the sample mean $\bar{x}$.

- The above confidence level then indicates that, **for the given $\bar{x}$ and $S_{\bar{x}}$, there is a 95% probability that the true mean (m_x) lies within the interval $[\bar{x} - d_{95}, \bar{x} + d_{95}]$.**

Confidence interval (CI) based on Gaussian and Student's t distributions

- When m_x is **unknown** and σ_x is **known**, with number of samples $n > 30$, define:

$$z^* = \Psi^{-1} \left(1 - \frac{\alpha}{2} \right) = -\Psi^{-1} \left(\frac{\alpha}{2} \right) \quad (112)$$

where $C = (1 - \alpha) \times 100\%$ is the confidence level, Ψ is the CDF of standard Gaussian. For example, when $\alpha = 0.05$, $C = 95\%$, $z^* = 1.96$. The CI is defined as:

$$\text{CI} = \left[\bar{x} - z^* \frac{\sigma_x}{\sqrt{n}}, \quad \bar{x} + z^* \frac{\sigma_x}{\sqrt{n}} \right] \quad (113)$$

- When m_x is **unknown** and σ_x is **unknown**, or when σ_x is **known** but $n \leq 30$, we use the (unbiased) sample standard deviation S_x to replace σ_x . Define:

$$t^* = F_{n-1}^{-1} \left(1 - \frac{\alpha}{2} \right) = -F_{n-1}^{-1} \left(\frac{\alpha}{2} \right) \quad (114)$$

where F_{n-1} is the CDF of Student's t distribution with $(n - 1)$ degrees of freedom. In statistics, t^* is also written as $t_{(n-1), \frac{\alpha}{2}}$. The CI is defined as:

$$\text{CI} = \left[\bar{x} - t^* \frac{S_x}{\sqrt{n}}, \quad \bar{x} + t^* \frac{S_x}{\sqrt{n}} \right] \quad (115)$$

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8. Normality Tests

8.1. Test of skewness coefficient

8.2. Shapiro-Wilk test for normality

- In addition to the use of the Student's t -distribution to obtain a confidence interval, one can employ tests for examining the normality of the estimated sample mean. This further improves confidence in the use of the CLT. If the sample mean is normally distributed, then **the skewness coefficient has to be zero** and, in the case of a finite sample size, should be smaller than a limit.
- The **skewness coefficient** C is determined by:

$$C = \frac{1}{S_x^3} \cdot \frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^3 = \frac{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^3}{\left[\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \right]^{\frac{3}{2}}} \quad (116)$$

- If C is larger than an upper limit, then the user has to make a decision on the use of the derived confidence level. There are two situations that may arise

- 1 The use of the confidence level is **doubtful** if

$$C^2 \geq (1.96)^2 \times \frac{6(N-1)(N-2)}{N(N+1)(N+3)} \quad (117)$$

- 2 The use of the confidence level is **meaningless** if

$$C^2 \geq (2.81)^2 \times \frac{6(N-1)(N-2)}{N(N+1)(N+3)} \quad (118)$$

Outline

8. Normality Tests

8.1. Test of skewness coefficient

8.2. Shapiro-Wilk test for normality

- The **Shapiro-Wilk test** is a test of normality in frequentist statistics. It was published in 1965 by Samuel Sanford Shapiro and Martin Wilk. It is also referred to as the **W -test**, determines a parameter W that should not exceed an expected value.
- It tests the the null hypothesis that a set of samples $x_1, x_2, \dots, x_N$ come from a normal distribution. The test statistic is defined as

$$W = \frac{\left(\sum_{i=1}^N a_i x_{(i)}\right)^2}{\sum_{i=1}^N (x_i - \bar{x})^2} \quad (119)$$

where $x_{(i)}$ is the i^{th} order statistic, i.e., the i^{th} smallest number in the sample set.

- The coefficients a_i are given by

$$(a_1, a_2, \dots, a_N) = \frac{\mathbf{m}^\top \mathbf{V}^{-1}}{\mathbf{C}}$$

- The vector $\mathbf{m} = [m_1, m_2, \dots, m_N]^\top$ is made of the expected values of the order statistics of i.i.d. RVs sampled from the standard normal distribution;
- $\mathbf{V}$ is the covariance matrix of those normal order statistics;
- $\mathbf{C}$ is a vector norm

$$\mathbf{C} = \|\mathbf{V}^{-1} \mathbf{m}\| = \sqrt{\mathbf{m}^\top \mathbf{V}^{-1} \mathbf{V}^{-1} \mathbf{m}}$$

- The procedure for estimation of W is given below:

- 1 For the sample size of N with outcomes $x_1, x_2, \dots, x_N$, compute the sample mean and sample variance.
- 2 Arrange $\{x_i\}_{i=1}^N$ in increasing order of magnitude, relabel them as y_i 's, and then compute the following summation

$$b = \sum_{i=1}^N a_i y_i \quad (120)$$

- 3 Compute w using the following formulation

$$W = \frac{b^2}{(N-1)S_x^2} \quad (121)$$

- Given N and W , one can estimate the probability of normality of the sample mean. If W has a small value $W < 0.806$, this probability is low (below 10%); while if $W > 0.986$, there is over 95% chance that the sample mean follows a normal distribution.
- In summary, both tests are straightforward, while the skewness test may require significantly less computing time. There are several other tests, such as the **Anderson-Darling test**, **Lilliefors test**, **Kolmogoro-Smirnov test**, **D'Agostino's K-squared test**, etc.